

Lec1

Introduction

Web development is the process of creating websites and web applications that run on the internet. It includes designing web pages, writing code, and managing databases to make websites interactive and user-friendly. Web development is mainly divided into three parts:

Frontend Development—Works on the visible part of the website using HTML, CSS, and JavaScript.

Backend Development—Handles server-side logic, databases, and authentication using languages like Python, PHP, Java, or Node.js.

Database Management—Stores and manages website data using MySQL, MongoDB, etc.

Example

A simple example of web development is a login page for a website.

The frontend creates the form where users enter their username and password.

The backend checks whether the entered details are correct.

The database stores the user information securely.

HTML

HTML(HypertextMarkupLanguage)usesamarkupsystemcomposedofelementswhic
hrepresentspecificcontent.MarkupmeansthatwithHTMLyoudeclarewhatispresented
toaviewer,nothowitispresented.VisualrepresentationsaredefinedbyCascadingStyle
Sheets(CSS)andrealizedbybrowsers.

HTML is sometimes called a programming language but it has no logic, so is a markup language. HTML tags provide semantic meaning and machine-readability to the content in the page.

An element usually consists of an opening tag (`<element_name>`), a closing tag (`</element_name>`), which contain the element's name surrounded by angle brackets, and the content in between:

```
<element_name>...content...</element_name>
```

There are some HTML elements that don't have a closing tag or any contents. These are called void elements. Void elements include `<img>`, `<meta>`, `<link>` and `<input>`.

Creating a simple page

HTML files can be created using any text editor. The files must be saved with a .html or .htm extension in order to be recognized as HTML files.

Once created, this file can be opened in any web browser.

Extension

Auto Close Tag

Auto Complete Tag

Auto Rename Tage

Live Server

Prettier

Tag	Meaning
<!DOCTYPE>	Defines the HTML version used in the document. In this case it is HTML5.
<html>	Opens the page and defines the primary language of the page.
<head>	Contains metadata, stylesheets, and scripts for the HTML document.
<meta>	Provides metadata about the document such as character encoding (UTF-8).
<title>	Defines the title of the webpage shown in the browser tab.
<body>	Contains all the visible content displayed to users.
<h1>	Represents a level 1 heading for the page.
<p>	Represents a paragraph of text.

Basic Tags

Headings

Paragraph

Image

Link

Heading

HTML provides not only plain paragraph tags, but six separate header tags to indicate headings of various sizes and thicknesses. Enumerated as heading 1 through heading 6, heading 1 has the largest and thickest text while heading 6 is the smallest and thinnest, down the paragraph level.

Defining a heading

<h1>Heading 1</h1>

<h2>Heading 2</h2>

<h3>Heading 3</h3>

<h4>Heading 4</h4>

<h5>Heading 5</h5>

<h6>Heading 6</h6>

Correct structure matters

Search engines and other user agents usually index page content based on heading elements, for example to create table of contents, so using the correct structure for headings is important.

In general, an article should have one h1 element for the main title followed by h2 subtitles—going down a layer if necessary. If there are h1 elements on a higher level they shouldn't be used to describe any lower level content.

Example

```
<h1>Main title</h1>
```

```
<p>Introduction</p>
```

```
    <h2>Reasons</h2>
```

```
        <h3>Reason 1</h3>
```

```
        <p>Paragraph</p>
```

```
        <h3>Reason 2</h3>
```

```
        <p>Paragraph</p>
```

```
    <h2>In conclusion</h2>
```

```
    <p>Paragraph</p>|
```

Paragraphs

The HTML `<p>` element defines a paragraph:

```
<p>This is a paragraph.</p>
```

```
<p>This is another paragraph.</p>
```

Display-

You cannot be sure how HTML will be displayed.

Large or small screens, and resized windows will create different results.

With HTML, you cannot change the output by adding extra spaces or extra lines in your HTML code.

The browser will remove any extra spaces and extra lines when the page is displayed:

```
<p>This is another paragraph,extra spaces will be removed by browsers</p>
```

Link

Anchor tags are commonly used to link separate webpages, but they can also be used to link between different places in a single document, often within table of contents or even launch external applications.

Parameter	Details
<code>href</code>	Specifies the destination address. It can be an absolute URL, relative URL, or the name of an anchor.
<code>hreflang</code>	Specifies the language of the linked resource using the <code>href</code> attribute.
<code>rel</code>	Defines the relationship between the current document and the linked document.
<code>target</code>	Specifies where to open the linked document, such as <code>_blank</code> , <code>_self</code> , <code>_parent</code> , or <code>_top</code> .
<code>title</code>	Provides extra information about the link, usually shown as tooltip text.
<code>download</code>	Specifies that the target file should be downloaded when the user clicks the hyperlink.

Example

Link to another site

```
<a href="http://example.com/">Link to example.com</a>
```

Link to an anchor

```
<a href="#Topic1">Click to jump to the First Topic</a>
```

Link to a page on the same site

```
<a href="/example">Text Here</a>
```

Link that dials a number

```
<a href="tel:1 1234567890">Call us</a>
```

Open link in new tab/window

```
<a href="example.com" target="_blank">Text Here</a>
```

Link that runs email client

```
<a href="mailto:example@example.com">Send email</a>
```

Images

To add an image to a page, use the image tag.

Image tags (img) do not have closing tags. The two main attributes you give to the imgtag are src, the image source and alt, which is alternative text describing the image.

```
<imgsrc="images/hello.png" alt="Hello World">
```

Parameters	Details
<code>src</code>	Specifies the URL of the image.
<code>srcset</code>	Defines images to use in different situations, such as high-resolution displays or small monitors.
<code>sizes</code>	Specifies image sizes between breakpoints.
<code>crossorigin</code>	Defines how the element handles cross-origin requests.
<code>usemap</code>	Specifies the name of the image map to use.
<code>ismap</code>	Indicates whether the image is a server-side image map.
<code>alt</code>	Provides alternative text if the image cannot be displayed.
<code>width</code>	Specifies the width of the image (optional).
<code>height</code>	Specifies the height of the image (optional).

Responsive image using the srcset attribute Using srcset

```

```

If view port is larger than 1200px, image is exactly 580px (for example our content is centered in container which is max 1200px wide. Image takes half of it minus margins).

If view port is between 640px and 1200px, image takes 48% of viewport (for example image scales with our page and takes half of view port width minus margins).

If viewport is any other size, in our case less than 640px, image takes 98% of viewport (for example image scales with our page and takes full width of viewport minus margins). Media condition must be omitted for last item.

Other Tags

Highlighting (mark)

Bold (strongorb)

Italic (emor i)

Underlined (u)

Abbreviation (abbr)

Inserted (ins)

Deleted (Del)

Stricken (s)

Superscript (sup)

Subscript (sub)

Homework

List

Anchor

Div

Span

Try all tags

Task

Create a basic HTML page with headings, paragraphs, images, and links